SOC L1 LAB · INC-004

INC-004 Detection Rules

Two rules for root persistence, and the legitimate admin work they also catch

INCIDENT	INC-004, Cron and systemd persistence rules
REPORT DATE	2026-11-01
CLASSIFICATION	Lab exercise, sanitized for publication
ENVIRONMENT	Wazuh 4.14.7 · Debian 12 · QEMU/KVM
SEVERITY	Critical

Independent portfolio lab. No production systems, client data, or real credentials.

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These are the two Wazuh rules added for INC-004. Both were loaded in 4.14.7 and tested against legitimate cron/systemd administration and a live root persistence run on 2026-10-31.

The deployed file is `rules/local_rules.xml`.

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1. Inventory

ID	Trigger	Level	MITRE
100129	FIM: file added/modified under `/etc/cron.d/`	12	T1053.003
100130	FIM: file added/modified under `/etc/systemd/system/`	10	T1543.002

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2. The rules

```
<group name="local,syscheck,">

  <rule id="100129" level="12">
    <if_sid>550,554</if_sid>
    <match>/etc/cron.d/</match>
    <description>FIM: cron file added or modified (possible persistence)</description>
    <mitre>
      <id>T1053.003</id>
    </mitre>
    <group>syscheck,persistence,</group>
  </rule>

  <rule id="100130" level="10">
    <if_sid>550,554</if_sid>
    <match>/etc/systemd/system/</match>
    <description>FIM: systemd unit or enable-link changed (possible persistence)</description>
    <mitre>
      <id>T1543.002</id>
    </mitre>
    <group>syscheck,persistence,</group>
  </rule>

</group>
```

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3. Rule-by-rule

100129: cron file added or modified

- **Parents**
  550 (modified) and 554 (added).
- **Match**
  The `/etc/cron.d/` path substring.
- **Tested**
  23:17:25.115 on `/etc/cron.d/updates`.
- **False positives**
  Every legitimate cron job. The baseline produced two hits for one legitimate file (add and modify).
- **Gap**
  User crontabs in `/var/spool/cron/crontabs` are not covered.

100130: systemd unit or enable-link changed

- **Parents**
  550 and 554.
- **Match**
  The `/etc/systemd/system/` path substring.
- **Tested**
  23:17:28.122 (`updates.service`), 23:17:28.128 (`updates.timer`), 23:17:31.309 (`timers.target.wants/updates.timer`).
- **False positive**
  Every `systemctl enable`, because it creates a symlink under `*.wants/`. The baseline produced one hit for enabling cron.
- **Why level 10**
  Enable events are frequent. The rule cannot distinguish a symlink from a new unit body with one `<match>`, so a lower level is the safer default.

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4. Test matrix

Action	Expected
Legit `/etc/cron.d/apt-maintenance` added	100129
Same file modified	100129
`systemctl disable cron`	silent (553)
`systemctl enable cron`	100130 on the enable symlink
Attack `/etc/cron.d/updates`	100129
`updates.service` written	100130
`updates.timer` written	100130
`systemctl enable --now updates.timer`	100130 on the enable symlink
`tee`, `systemctl`, `id`	80792 command trail
`sudo` writes	5402

Baseline counts: 2× 100129, 1× 100130. Attack counts: 1× 100129, 3× 100130.

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5. Coverage gaps

- **User crontabs**
  `/var/spool/cron/crontabs` is not monitored. A user-level job is invisible to 100129.
- **Package units**
  Units in `/lib/systemd/system` are outside the FIM path. Only the `/etc` overrides and enable links are covered.
- **Deletion**
  100129/100130 use 550/554. A deletion is 553, so removing persistence is silent.
- **Symlink vs body**
  100130 cannot distinguish a legit `enable` link from a new unit file. Both are level 10.
- **Intent**
  Neither rule knows whether a change was approved. Change control is the only discriminator.
- **Execution**
  Neither rule proves the job or service ran. Audit and the proof file do.

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6. Tuning

6.1 Split 100130

Best next step:

- level 12 for a new `.service` / `.timer` body.
- level 5 for a `*.wants/` enable link.
- correlate the body + enable + execution for a Critical sequence.

This requires two rules with different matches (one on `.service`, one on `.timer`, or a path allow-list) and a way to exclude `*.wants/`. Since `<match>` is a substring and only one match per rule is reliable, use separate rules and test the precedence.

6.2 Cover user crontabs

Add FIM/audit for `/var/spool/cron/crontabs` and the `crontab` command. Keep the rule at level 12 for changes, with a change-ticket check.

6.3 Deletion rules

Add `if_sid 553` rules for the same paths. Removing a unit or cron file is anti-forensics. It should not be silent.

6.4 Change-control integration

Match every hit to an approved change, a package transaction, or a configuration management run. Do not suppress the rule. Suppress the *known* deployment ID.

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7. Deployment

```
cd "$HOME/wazuh-docker/single-node"
docker compose up -d wazuh.manager
sleep 20

for id in 100129 100130; do
  printf '%s: ' "$id"
  docker exec single-node-wazuh.manager-1 \
    grep -c "id=\"$id\"" /var/ossec/etc/rules/local_rules.xml
done
```

Validate 100129 with a disposable cron file and 100130 with a `systemctl enable` action. Run the legitimate baseline before any attack so the false positives are recorded.

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8. MITRE ATT&CK

Technique	ID	Rule
Scheduled task/job: cron	T1053.003	100129
Create or modify system process: systemd service	T1543.002	100130
Abuse elevation control: sudo	T1548.003	context (80792/5402)
Valid accounts	T1078	context (actor)

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9. Caveats

- Tested on Debian 12, systemd 252, cron 3.0, Wazuh 4.14.7.
- `/etc` must be in FIM realtime.
- A host using another init system needs different coverage.
- The rules detect file change, not execution or authorization.